

MOSSEL BAY, South AfricaWMO: **689280**Lat: **34.190S**Lon: **22.132E**Elev: **194**StdP: **14.59**Time Zone: **2.00 (E02)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	46.3	47.9	31.8	26.3	64.9	35.2	30.3	62.0	29.3	59.3	23.1	60.1	7.9	290	0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.7	81.1	64.7	78.2	66.3	76.3	66.6	71.6	76.0	70.4	74.7	69.3	73.6	7.6	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.0	111.2	74.1	68.8	106.7	73.2	67.7	102.5	72.3	35.5	76.4	34.5	74.8	33.6	73.8	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.9	20.9	17.2	DB	44.1	91.6	0.9	4.5	43.4	94.8	42.9	97.5	42.3	100.0	41.7	103.3	(4)
(5)			WB	40.3	73.7	1.0	1.5	39.6	74.8	39.0	75.7	38.5	76.5	37.8	77.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.4	69.6	69.4	67.5	64.1	62.1	58.9	58.2	59.4	61.9	64.1	67.6	(6)	
(7)		DBStd	5.58	2.78	2.86	3.14	4.45	4.19	4.28	4.94	3.98	4.10	3.40	3.04	3.26	(7)
(8)		HDD50	1	0	0	0	0	0	0	1	0	0	0	0	0	(8)
(9)		HDD65	1172	1	2	11	64	112	192	223	217	176	109	52	13	(9)
(10)		CDD50	4881	607	543	543	424	374	266	254	253	282	368	422	545	(10)
(11)		CDD65	577	143	125	89	38	21	8	11	5	9	11	24	93	(11)
(12)		CDH74	1144	193	175	120	158	116	47	58	40	68	36	19	114	(12)
(13)		CDH80	181	5	14	13	48	31	7	8	9	27	10	3	6	(13)
(14)	Wind	WSAvg	7.5	8.0	7.7	7.1	6.6	6.3	7.5	7.5	7.6	7.6	8.0	8.3	8.0	(14)
(15)	Precipitation	PrecAvg	14.9	1.0	0.9	1.1	1.6	1.3	1.2	1.1	1.5	1.1	1.6	1.3	1.0	(15)
(16)		PrecMax	28.3	6.7	3.3	2.8	5.4	5.1	4.1	3.5	6.6	5.5	4.8	6.2	2.8	(16)
(17)		PrecMin	7.2	0.0	0.0	0.1	0.1	0.1	0.1	0.1	0.3	0.0	0.2	0.1	0.0	(17)
(18)		PrecStd	3.9	1.2	0.6	0.7	1.3	1.1	0.7	0.8	1.5	1.0	1.2	1.1	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.0	81.5	80.9	86.0	83.8	80.6	80.2	79.9	82.6	80.8	76.0	79.9	(19)
(20)			MCWB	71.0	71.6	67.7	64.9	61.6	57.5	57.8	57.7	60.4	64.6	65.9	69.8	(20)
(21)		2%	DB	77.8	77.7	76.8	77.9	76.1	73.9	74.4	70.5	72.0	72.0	72.8	76.9	(21)
(22)			MCWB	70.1	69.7	67.6	62.5	59.5	55.3	55.7	56.1	58.9	63.0	64.6	69.1	(22)
(23)		5%	DB	76.1	76.0	74.6	73.0	70.8	69.4	69.5	66.6	67.3	69.1	71.0	75.0	(23)
(24)			MCWB	69.1	68.7	67.2	62.8	58.7	54.6	54.9	56.1	58.8	62.0	63.5	67.8	(24)
(25)		10%	DB	74.5	74.3	72.7	69.9	67.6	66.0	65.5	63.8	65.0	67.2	69.5	73.3	(25)
(26)			MCWB	68.2	68.0	66.5	62.8	58.8	55.0	54.8	56.1	58.2	61.2	62.8	66.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.0	73.6	71.2	68.2	64.3	61.3	60.7	61.9	64.5	66.7	68.5	72.1	(27)
(28)			MCDWB	77.4	79.1	76.4	76.8	71.9	67.2	67.4	67.0	72.4	71.7	73.8	77.1	(28)
(29)		2%	WB	71.6	71.5	69.5	66.4	62.7	59.5	59.0	60.2	62.1	64.7	66.6	70.1	(29)
(30)			MCDWB	75.8	75.6	74.0	72.0	68.9	65.3	65.9	64.1	66.8	69.3	70.9	75.0	(30)
(31)		5%	WB	70.4	70.1	68.5	65.0	61.7	58.0	57.8	58.9	60.7	63.4	65.0	68.9	(31)
(32)			MCDWB	74.3	74.3	72.8	69.7	67.0	64.0	64.1	63.4	65.5	67.8	69.1	73.6	(32)
(33)		10%	WB	69.4	69.0	67.4	63.9	60.7	56.8	56.7	57.6	59.3	62.2	63.8	67.8	(33)
(34)			MCDWB	73.2	73.1	71.5	68.3	65.3	62.9	62.8	62.1	64.1	66.2	68.1	72.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.7	8.9	9.0	10.6	11.6	12.5	12.4	11.8	10.9	9.6	9.4	9.2	(35)
(36)			MCDBR	10.3	11.4	11.9	17.0	17.8	18.3	18.7	17.0	15.0	12.3	11.4	11.3	(36)
(37)			MCWBR	5.6	6.1	6.1	7.4	7.5	8.0	7.9	7.8	7.6	6.5	6.4	6.1	(37)
(38)		5% WB	MCDBR	8.7	9.7	10.2	12.4	12.7	13.2	13.5	12.6	12.4	10.0	10.1	10.0	(38)
(39)			MCWBR	5.1	5.7	5.7	6.6	6.9	7.4	7.7	7.1	7.1	5.9	6.3	5.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.371	0.374	0.360	0.336	0.322	0.310	0.312	0.339	0.351	0.372	0.360	0.368	(40)	
(41)		taud	2.500	2.496	2.524	2.574	2.591	2.596	2.584	2.502	2.450	2.425	2.495	2.495	(41)	
(42)		Ebn at Noon	306	299	291	281	268	264	269	275	289	296	308	309	(42)	
(43)		Edh at Noon	36	35	32	27	23	22	23	28	34	37	36	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2371	2120	1732	1287	961	823	903	1164	1582	1939	2285	2457	(44)	
(45)		RadStd	86	71	82	56	50	47	51	48	75	103	98	133	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (5 neighbors)		N/A	N/A	+3.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page